CADY H. NGO

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EDUCATION

B.A. in Mathematics and Economics (*with honors*), minor in Statistics 2025
Grinnell College

- Relevant Coursework: Abstract Algebra, Applied Game Theory, Applied Statistics, Biological Psychology, Data Science, Differential Equations, Design & Analysis of Experiments, Econometrics, Number Theory, Probability Theory, Real Analysis, Research Methods, Statistical Modeling

Visiting Honours Student 2024
University of Edinburgh

- Relevant Coursework: Computing & Numerics, Programming & Numerical Methods for Economics

RESEARCH EXPERIENCE

Chen Predoctoral Fellow in Social and Decision Neuroscience 2025–Present
Division of the Humanities and Social Sciences & Chen Institute for Neuroscience, Caltech

- PI: Colin Camerer
- Developing multi-agent LLM trading infrastructure to simulate human-like bubble behavior (via NLP extraction, LLM self-prompting, and reinforcement learning) toward AI substitution for human participants
- Administering fMRI tasks manipulating curiosity-risk tradeoffs via calibrated electrical stimulation to examine striatum-amygdala competition in human decision-making under uncertainty

Visiting Research Fellow 2024–2025
Division of the Humanities and Social Sciences, Caltech

- PI: Colin Camerer
- Conducted in-lab and virtual experiments via Caltech's SSEL and Prolific, collecting experimental stock market and biometric data (e.g., EDA, HR, BVP, skin temperature) using Empatica wearables
- Implemented machine learning (e.g., LASSO, RFs) and forecasting models to predict asset bubble formation
- Pioneered project using affect-based photo databases to induce exogenous insular cortex activation

Mentored Advanced Project in Computational Modeling of Music 2025
Department of Psychology & Department of Music, Grinnell College

- Self-developed an original interdisciplinary framework integrating game theory, statistical learning, and music cognition to model listener expectation and musical preference in Western classical music
- Designed and analyzed normal form, sequential, and signaling games to formalize strategic interactions between melody and bass under coordination and asymmetric information
- Grounded theoretical payoffs in entropy-based measures derived from Markov models and PCA summaries of pitch transition data from Beethoven piano sonatas

Research Fellow at Opportunity Insights 2023
Department of Economics, Harvard University

- PIs: Raj Chetty and Gregory Bruich
- Employed Stata to clean, sort, and analyze census data sets, investigating the correlation between highways and declining social mobility in major American cities
- Designed empirical models (i.e., DID, RD), conducted lit review, and produced paper with replication code

WORKING PAPERS

Ngo, C. H.[†], Henning, T.[†], Spoon, R. L.[†], Toma, F. M.[†], Han, J., Smith, A., & Camerer, C. F. Skin in the Game: Electrodermal Arousal, Expectations, and Bubbles in Experimental Asset Markets. *Under review.* [Paper] [Poster]

Ngo, C. H. Feeding Attendance: The Impact of California's Universal Meals Program on Public School Absenteeism. [Paper] [Poster] [Slides]

Zhang, B., Towner, E., **Ngo, C. H.**, & Mobbs, D. Neural Mechanisms of Curiosity in Risky Decision-Making: An fMRI Investigation. [Github]

SELECTED WORKS IN PROGRESS

Ngo, C. H., Manning, B. S., Avery, C. N., & Camerer, C. F. Persona Construction for Large Language Model Agents in Experimental Asset Markets. [Paper] [Poster]

Ngo, C. H. & Camerer, C. F. Ultra-Processed Foods, Supernormalization, and Rational Inattention.

†: *equal contribution*

AWARDS & GRANTS

External

Midwest Economics Association's Best Undergraduate Paper, Honorable Mention	2025
Thomas and Charlotte Taylor Scholarship ($10,000)	2024–2025
San Gabriel Women's Club Scholarship ($4,000)	2021–2025
PREDOC.org Service Honorarium ($500)	2024
National Institute of Statistical Sciences Ingram Olkin Forums Travel Grant ($450)	2023
PREDOC.org Summer Research Award ($2,000)	2023

Caltech

Tianqiao and Chrissy Chen Predoctoral Fellowship Grant ($2,500)	2025

Grinnell

Gilbert DeBartolo '69 Memorial Prize in Economics ($1,300)	2025
Dean's Office Research Conference Travel Grant ($1,920)	2025
Professional Development Grant ($1,700)	2024–2025
Founder's Scholarship ($64,000)	2021–2025
Grinnell Choice Scholarship ($40,000)	2021–2025
Graduate School Scholarship ($700)	2024
Summer Research Grant ($4,000)	2024
Externship Grant ($960)	2023
Hill Distinguished Award for Outstanding Music Performance ($750)	2023

CONFERENCE & WORKSHOP PRESENTATIONS

2026 Conference on Language Modeling (COLM): Workshop on Agent Behavior in San Francisco, CA
Poster: Persona Construction for Large Language Model Agents in Experimental Asset Markets

Research and Behavioral Finance Conference (RBFC) at VU Amsterdam
Talk: A Recipe for Creating Human-Like Bubbles in LLM-Populated Experimental Asset Markets

Economic Science Association (ESA) World Meeting in Los Angeles, CA
Talk‡: A Recipe for Creating Human-Like Bubbles in LLM-Populated Experimental Asset Markets

2nd Workshop on AI Behavioral Science at Stanford
Talk‡*:* Persona Prompting of LLMs to Match Human Behavior in Experimental Asset Markets

Behavioral and Experimental Economics Coffee Hour at Caltech
Talk‡*:* Psychobiological Correlates of Experimental Asset Bubbles

Chen Institute Neuro Lunch Meeting at Caltech
Talk‡*:* Psychobiological Correlates of Experimental Asset Bubbles

2025 15th Annual Behavioral Science and Health Symposium at University of Pennsylvania
Talk: Ultra-Processed Foods, Supernormalization, and Rational Inattention

PREDOC.org Research Conference at the World Bank in Washington, D.C.
Poster: The Impact of California's Universal Meals Program on Public School Absenteeism

Midwest Economics Association 89th Annual Meeting in Kansas City, MO
Talk⁕*:* The Impact of California's Universal Meals Program on Public School Absenteeism

2024 Research in Behavioral Finance Conference (RBFC) at VU Amsterdam
Talk‡*:* The Psychobiological Predictability of Experimental Asset Bubbles

Los Angeles Experiments (LAX) Workshop at Caltech
Poster: The Market's Hidden Signals: Using Biometric Data to Predict Financial Bubbles

SFP Summer Seminar Day at Caltech
Poster: The Market's Hidden Signals: Using Biometric Data to Predict Financial Bubbles

2023 Undergraduate Research Conference at the Federal Reserve Board of Governors in Washington, D.C.
Poster: America's Interstate System as an Agent of Dampening Social Mobility

PREDOC.org Seminar Series
Talk: America's Interstate System as an Agent of Dampening Social Mobility

‡: *presented by co-author* ⁕: *special recognition or award*

TEACHING & SERVICE

Teaching

TA for Colin Camerer, AI and Social Sciences (EC/PSY/SS 104), Caltech — Fall 2026

Course Tutor, Research Methods (PSY 225), Grinnell College — Fall 2024, Spring 2025

Service

Summer Undergraduate Research Fellowship (SURF) Mentor, Caltech — 2026

Board Member and Mentor, Diversity and Inclusion Club for Economics (DICE), Grinnell College — 2024–2025

Elected Board Member, Student Educational Policy Committee for Economics, Grinnell College — 2022–2025

Advisory Board, PREDOC.org — 2023–2025

Ad Hoc Reviewing

PNAS Nexus

AFFILIATIONS

Caltech Resnick High Performance Computing Center (HPC) — 2026–Present

Caltech Brain Imaging Center (CBIC) — 2025–Present

Caltech Social Science Experimental Laboratory (SSEL) — 2024–Present

CERTIFICATES & CERTIFICATIONS

Certificates

Math for AI, Division of Engineering and Applied Sciences, Caltech 2025

Using Big Data to Solve Economic and Social Problems, Opportunity Insights and PREDOC.org 2023

Certifications

CITI Biomedical Research

CITI Hybrid Research

CITI Social & Behavioral Research

OTHER EXPERIENCE

Workshops Attended

Theory-Driven Experiments, Center for Theoretical and Experimental Social Sciences, Caltech 2026

Statistical Challenges in the Analysis of Police Use of Force, NISS Ingram Olkin Forum S^3 Series 2023

Selective Doctoral Preview Programs

Doctoral Preview Day, Brown University 2025

Colorado Advantage Program, University of Colorado Boulder 2025

Explore Michigan Engineering Research & Graduate Education (EMERGE), University of Michigan 2025

Employment

Research Assistant, Department of Economics, East Los Angeles College 2019

SKILLS

Programs

Python, R, Julia, Stata, LaTeX, SPSS

Experiment Software

Gorilla, oTree, Prolific, Qualtrics, Unity, WebGazer.js

Tools

Siemens Prisma 3 Tesla MRI, Empatica E4 & Embrace

MISC

Languages

English (native), Vietnamese (intermediate), Cantonese (elementary)

Citizenship

USA

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